



OBJECT ORIENTED ANALYSIS AND DESIGN

A study material for the students of GLS University

Session 1 — Why OOAD?

Importance, Industry Relevance & Course Overview

Session Objectives

- What is OOAD?
- Why is this subject included in MSc IT?
- How is OOAD used in the software industry?
- Why do companies design software before coding?
- What will they learn throughout the semester?
- How does OOAD help in interviews and real projects?

The Blueprint Analogy

SCENARIO 1

"Can an architect start constructing a building without a blueprint?"

SCENARIO 2

"Can a team of 50 developers start building Flipkart or Instagram directly by coding?"

Just as civil engineers create a blueprint before construction, software engineers create a design before development. **OOAD is the process of creating that software blueprint.**

Why Has This Subject Been Added in MSc IT?

Most students focus only on:

- Programming Languages
- Databases
- Web Development
- Mobile Apps

But in industry, coding is only one part of software development.

Companies need professionals who can:

- Understand client requirements
- Analyze business problems
- Design software solutions
- Communicate ideas with team members
- Create documentation before coding

The purpose of this subject: “How to think like a Software Engineer rather than only a Programmer.”

Real Industry Problem

Client says: “I want a Student Feedback Management System.”

Immediately, questions arise:

- Who will use it — students, faculty, admin?
- What reports are required?
- How is feedback stored?
- Can feedback be edited?
- Should login be required?

Without analysis → Result:

- Confusion
- Requirement changes
- Bugs
- Rework
- Project delays

Industry solution — before coding:

1. Understand requirements
2. Analyze the system
3. Design the solution
4. Then start development

What Exactly Is OOAD?

OOAD is the process of understanding a problem and designing a software solution before writing code.

Analysis (Understanding) -

- What problem needs to be solved?
- Who are the users?
- What are the requirements?

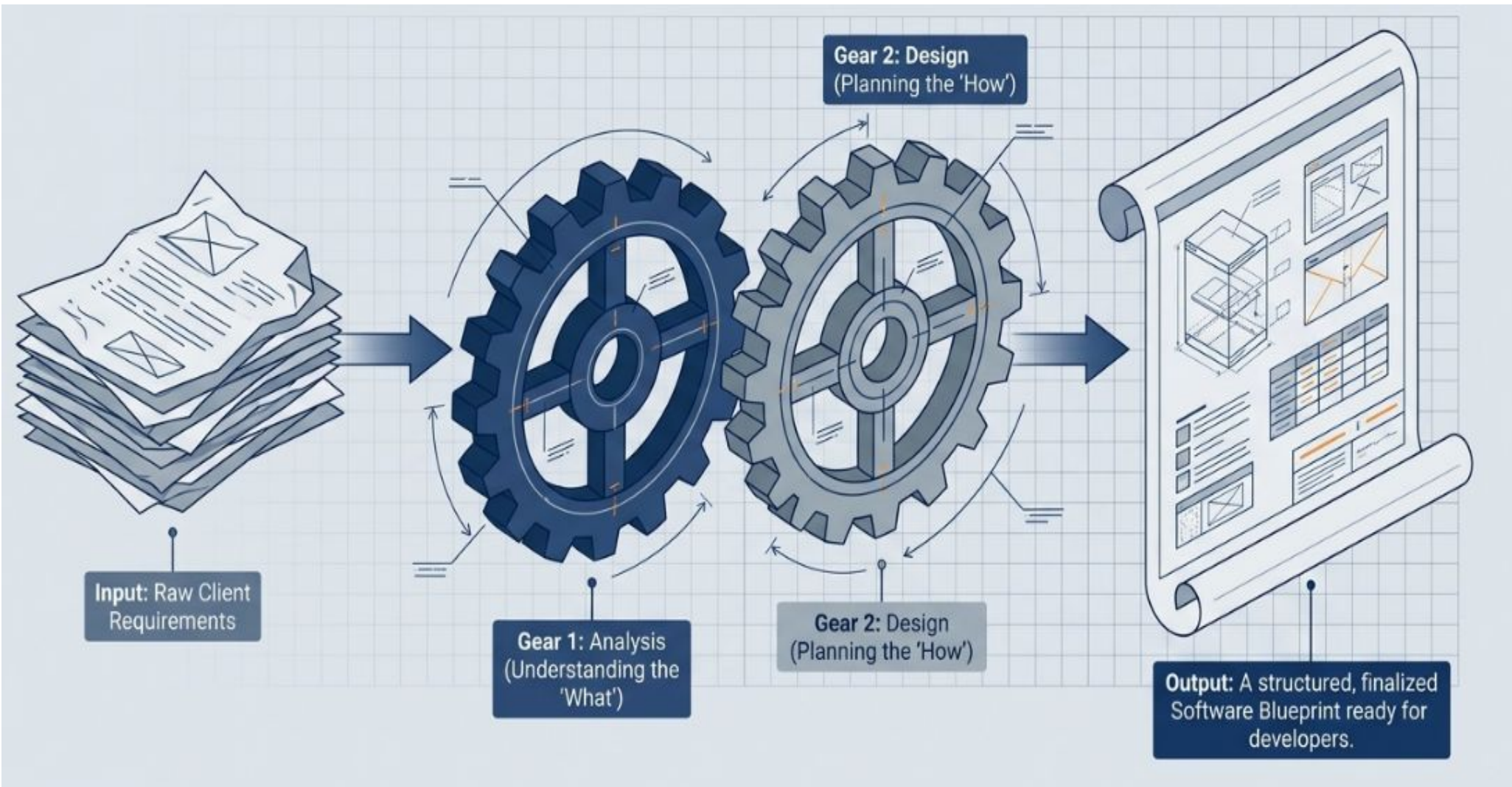
Example: Online Food Delivery App

Analysis identifies: Customer ·
Restaurant · Delivery Partner ·
Admin

Design (Planning how the solution will work) -

Classes
Modules
Relationships
Workflows

The engine of software architecture



Where OOAD Fits in Software Development



OOAD primarily focuses on the Analysis and Design phases — the critical steps before a single line of code is written. This is where requirements are clarified, system structure is planned, and potential problems are identified early.

Industry Importance of OOAD

Better Communication - Developers, testers, managers and clients understand the same design.

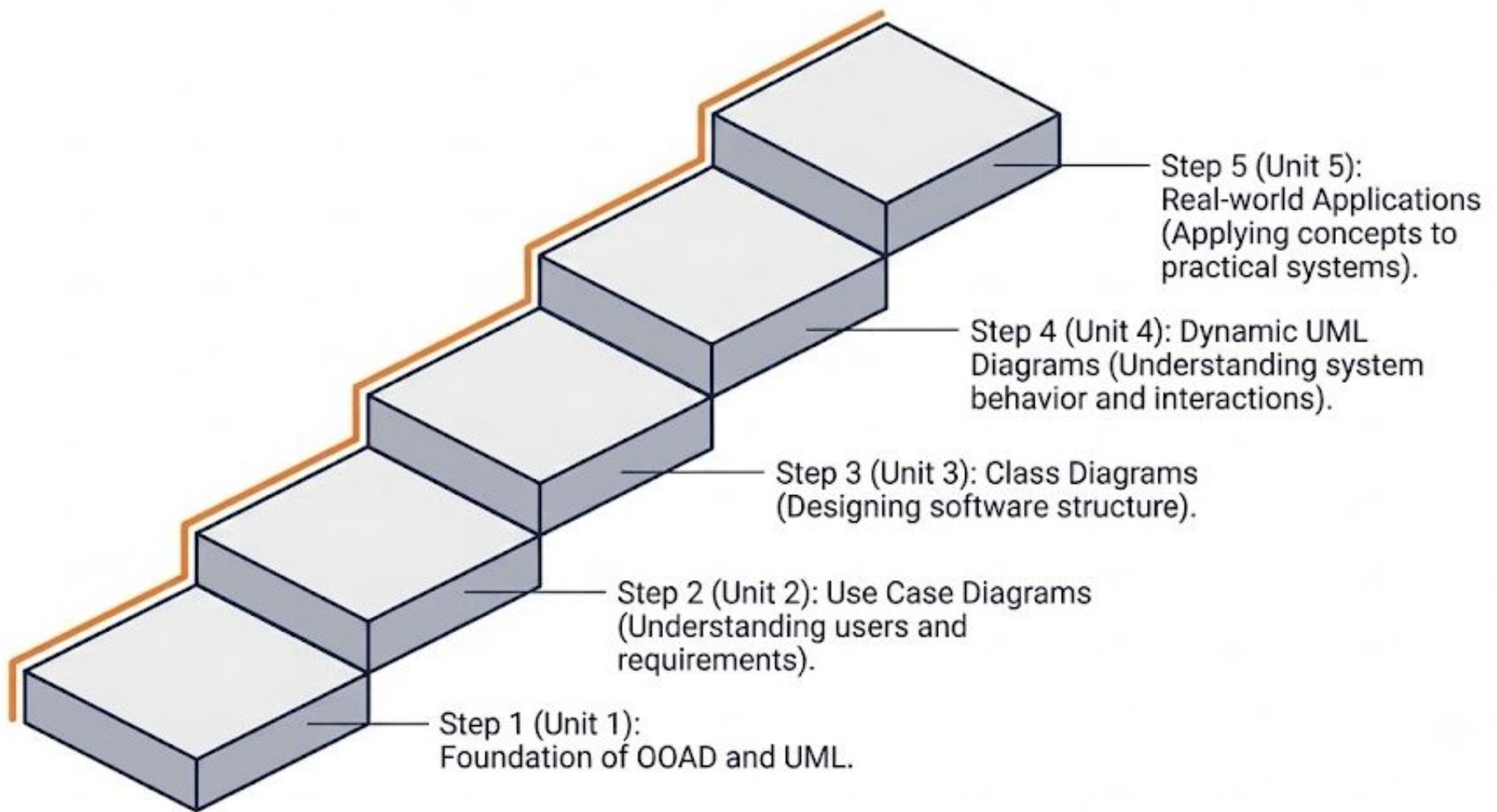
Reduces Errors - Problems are identified before coding starts.

Faster Development - design means less confusion.

Easier Maintenance - Future developers can understand the system.

Team Collaboration - Large teams can work on different modules efficiently.

What We Will Learn in This Subject



How This Subject Helps in Placements

Many interview questions are based on:

- UML
- use case
- class diagram
- inheritance
- object oriented concepts
- system design basics

Typical interview / design questions:

- Design a Library Management System
- Design an ATM System
- Design an Online Shopping System
- Design a Food Delivery App

Thank You